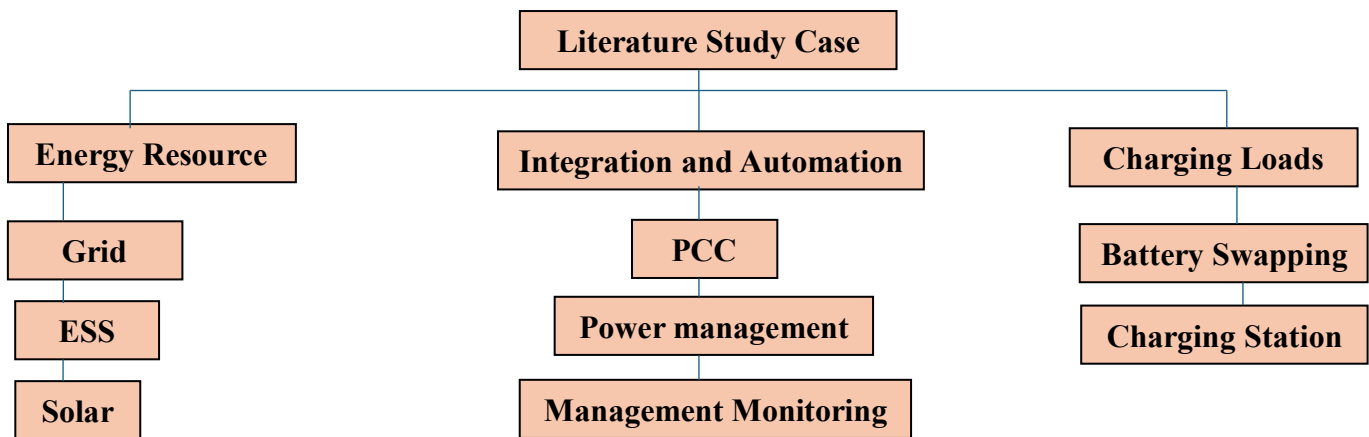


Chapter -2

LITERATURE REVIEW



The integration of IoT and renewable energy into EV charging infrastructure has been extensively explored to address grid stability, energy efficiency, and user convenience. Studies such as Solar monitoring ([1], [2]) and ESS optimization ([3], [14]) address energy harvesting and storage. Battery swapping logistics ([7]– [9]) mitigate grid strain through optimized station planning and mobile units. IoT-driven solutions ([4]– [6], [10]– [13], [16]) enhance microgrid operations, fault detection, and sensor placement, while residential charging studies ([11], [12]) tackle accessibility and scheduling. Power quality ([6], [15]) and safety innovations ([15], [16]) prioritize grid reliability and battery management. Despite advancements, gaps persist in hybrid system standardization (solar/ESS/grid at PCC) and economic viability. This work bridges these gaps by unifying IoT, predictive analytics, and dynamic energy distribution ([10], [12]), advancing scalable and cost-efficient EV infrastructure.

This literature case study underlines that integrating IoT with advanced analytics, sensor networks, and energy storage management can revolutionize EV charging infrastructures. Real-time data acquisition, intelligent scheduling, renewable integration, and adaptive fault detection together form a solid foundation for developing a next-generation, scalable battery charging management system.

Literature Survey Detailed

Paper Title	Author's, Publication & Year	About Paper	Outcome of Paper Understanding	Approach for My Project
A Pitaya Dye-Sensitized Solar Cell Monitor for Environmental Sunlight Intensity Detection	Cheng-Ta Chiang; Liang-Yu Chien - " IEEE Sensors Journal, 19-11, 03 February 2019	Explores techniques for monitoring solar cell performance and sunlight intensity to support renewable integration.	Underlines the importance of accurate solar monitoring in optimizing ESS capacity planning.	Integrate real-time solar monitoring tools within the IoT system to dynamically adjust charging based on available renewable energy.
Eternal-Thing: A Secure Aging-Aware Solar-Energy Harvester Thing for Sustainable IoT	Saswat Kumar Ram; Sauvagya Ranjan Sahoo; Banee Bandana Das; Kamalakanta Mahapatra; Saraju P. Mohanty - IEEE Transactions on Sustainable Computing, 6-2, 15 April 2020.	Combines energy harvesting with security and aging analysis to support sustainable IoT applications in solar energy.	Stresses power conditioning, MPPT, and long-term operational reliability.	Implement secure, aging-aware energy harvesting modules in the IoT architecture to ensure sustained battery charging performance under variable conditions.
ESS Simulation based on PV Generation Prediction	Tai-Yeon Ku et al., ICTC, 2018	Simulates the performance of an Energy Storage System (ESS) based on photovoltaic (PV) generation forecasts.	Demonstrates that predictive analytics can optimize ESS efficiency and renewable integration.	Apply predictive models to optimize the ESS integration in the charging management system and to balance renewable input with load demand.
IoT-based optimization of microgrid operation	Abdelrahman Ahmed et al., MEPCON, 2023/2024	Explores IoT applications to optimize microgrid operations, focusing on fault detection and energy efficiency improvements.	Shows real-time monitoring can reduce operational costs by dynamically responding to grid conditions.	Incorporate IoT-driven fault detection and microgrid optimization techniques to ensure reliable battery charging operations and grid stability.
An Efficient LoRa-Enabled Smart Fault Detection and Monitoring Platform for the Power Distribution System Using Self-Powered IoT Devices	George Y. Odongo et al., IEEE Access, 2022	Focuses on rapid fault detection in power distribution using a LoRaWAN-based platform with self-powered sensors.	Achieves fault detection within a short time frame (~100 ms) enhancing grid reliability.	Deploy LoRaWAN-based sensors in the IoT architecture to enable rapid fault detection and preventive maintenance in battery charging stations.
Development of IoT-Based Portable Power Quality Monitoring on Microgrids by Enhancing Protection Features	Isa Hafidz et al., IEEE Access, 2023	Examines portable devices for power quality monitoring that protect microgrid equipment from voltage fluctuations and harmonics.	Demonstrates enhanced protection and extended equipment life through continuous monitoring of power quality.	Implement portable IoT monitoring modules within the charging system to safeguard battery health and optimize power quality.
Optimal Planning of Battery Swapping and Charging Stations for the Urban Cities	Dan Zheng; Fushuan Wen; Jiansheng Huang - (SUPERGEN 2012), 04 April 2013.	Explores integration of battery swapping and charging stations with optimization techniques using mathematical programming.	Demonstrates that combining both approaches can reduce grid strain and enhance operational efficiency.	Design hybrid models within the IoT framework to enable seamless integration of battery swapping with charging management for dynamic load balancing.
Battery Swapping Mobile Service Station for Electric Vehicles Application	Bansilal Bairwa; Mrinal Sarvagya; Sanjeeva Kumar; Arpita Banik - 2021 Second International Conference on Smart Technologies in Computing, Electrical and Electronics (ICSTCEE), 15 February 2022.	Proposes a mobile battery swapping service station, focusing on dynamic architecture and real-time control of mobile units.	Differentiates battery swapping from traditional charging by emphasizing rapid service and operational flexibility.	Integrate mobile battery swapping unit data into the IoT platform for real-time monitoring and flexible energy management in multi-site deployments.
Optimal dispatch of electric vehicle batteries between battery swapping stations and charging stations	Xian Zhang; Guibin Wang - 2016 IEEE Power and Energy Society General Meeting (PESGM), 14 November 2016.	Discusses logistics challenges and the need for efficient battery dispatch between swapping and charging stations.	Identifies issues related to long charging times and grid load imbalances.	Develop a hybrid dispatch algorithm within the IoT architecture to optimize battery logistics and reduce grid strain during peak periods.
IoT Solutions for Electric Vehicles (EV) Charging Stations: A Driving Force Towards EV Mass Adoption	Sayantika Sikdar; Madhavi Damle, IIHC, 2023	Discusses EV growth, range anxiety issues, and how IoT can enhance smart charging station development and security.	Underlines the role of continuous monitoring and intelligent scheduling to prevent battery damage.	Incorporate IoT-enabled intelligent scheduling and real-time monitoring to optimize battery charging and enhance security.
Residential Electric Vehicle Charging Station Using the Internet of Things	Ashutosh Vijay Panchbhai et al., PEEIC, 2023/2024	Addresses challenges in home charging infrastructure; emphasizes the importance of user-friendly interfaces, accessibility, and flexible deployment.	Shows that IoT can enhance residential charging systems by simplifying user access and scheduling.	Develop an intuitive, user-friendly IoT interface and flexible scheduling module for residential battery charging management.
IoT enabled E-Vehicle Charging System with Battery Monitoring and Charge Scheduling	S. Arulmozhi et al., ICPEEV, 2023/2024	Proposes an energy-efficient strategy with real-time battery monitoring and dynamic charge scheduling.	Confirms that IoT-driven monitoring improves energy efficiency and reduces grid dependency.	Implement predictive scheduling algorithms integrated with real-time battery monitoring to manage charging operations optimally.
A Novel Sensor Placement Strategy for an IoT-Based Power Grid Monitoring System	Joe-Air Jiang; Jen-Cheng Wang; Hung-Shuo Wu; Chien-Hsing Lee; Cheng-Ying Chou; Li-Cheng Wu - IEEE Internet of Things Journal, 7-8, 30 April 2020.	Proposes optimized sensor placement for effective power grid monitoring and energy management, essential for calculating optimal ESS capacities.	Provides insights into how optimal sensor placement can enhance the accuracy of data acquisition.	Apply an optimized sensor placement strategy to improve the precision of battery parameter measurements and system performance analysis.
State-of-Charge Equalizations Comparison in a Multi-prosumer Environment Based on Real-Time Energy Management Systems with IoT-Support	Enrique A. Sanabria-Torres; Adriana C. Luna; Nelson L. Diaz; Fabio Andrade-Rengifo - 2023 IEEE Applied Power Electronics Conference and Exposition (APEC), 31 May 2023.	Focuses on balancing battery state-of-charge (SOC) in complex energy environments using real-time IoT monitoring and control.	Highlights the need to consider ESS characteristics and operational constraints for effective SOC management.	Integrate SOC equalization algorithms into the IoT framework to ensure uniform battery health and extend battery life during charging cycles.
IoT Based Electrical Vehicle Battery Management System with Charge Monitor and Fire Protection	Enrique A. Sanabria-Torres; Adriana C. Luna; Nelson L. Diaz; Fabio Andrade-Rengifo	Focuses on large-scale Li-ion battery management using IoT, cloud computing, and advanced SOC estimation methods.	Presents solutions for real-time monitoring, adaptive SOC estimation, and fire protection.	Embed cloud-based battery monitoring and adaptive SOC estimation within the IoT architecture to ensure safe and efficient charging management.
An IoT -based Power Management System for EV Charger	Mersin, Türkiye - 6th Advanced Engineering Days (AED), 5 March 2023.	Discusses the integration of sensors and communication modules (e.g., Arduino, Raspberry Pi) for monitoring and control in EV chargers.	Highlights the use of simple microcontroller platforms for data exchange and display in EV charging systems.	Utilize robust IoT communication protocols to enhance data accuracy and control across battery charging units.
Power Quality Enhancement in Integrated Grid for EVs Charging Stations Using Active Filter	Santosh Kumar Tripathi et al., PARC, 2024	Examines how EV charging affects power quality by causing harmonic distortion and proposes active filter techniques to mitigate these effects.	Demonstrates that active filters can significantly reduce total harmonic distortion (THD).	Integrate active filter solutions into the IoT architecture to ensure stable and high-quality power supply at the Point of Common Coupling (PCC).